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Criterio de Dini

Proposición 1 (Criterio de Dini). *Sea* $f \in \mathcal{L}^1([0,1))$ *1-periódica y* $x_0 \in \left(-1/2, 1/2\right)$ *tal que* $f(x_0) < \infty$*. Entonces,*

$$\exists \delta > 0 : \int_{|t|<\delta} \left| \frac{f(x_0+t) - f(x_0)}{t} \right| \mathrm{d}t < \infty \implies \lim_{N\to\infty} S_N(f)(x_0) = f(x_0).$$

***Demostración*:** Puesto que $\int_{-1/2}^{1/2} D_N(t)\,\mathrm{d}t = 1$ para todo $N \in \mathbb{N}$, tenemos que

$$\begin{aligned} S_N f(x_0) - f(x_0) &= \int_{-\frac{1}{2}}^{\frac{1}{2}} f(x_0 - y) D_N(y)\,\mathrm{d}y - f(x_0) \int_{-\frac{1}{2}}^{\frac{1}{2}} D_N(y)\,\mathrm{d}y \\ &= \int_{-\frac{1}{2}}^{\frac{1}{2}} \left[f(x_0-y) - f(x_0)\right] D_N(y)\,\mathrm{d}y \\ &\overset{\text{lem-formula-nucleo-dirichlet/1}}{\overset{\downarrow}{=}} \int_{-\frac{1}{2}}^{\frac{1}{2}} \left[f(x_0-y) - f(x_0)\right] \frac{\sin\left((2N+1)\pi y\right)}{\sin(\pi y)}\,\mathrm{d}y. \end{aligned}$$

Definimos $I_N f(x_0)$ como la integral anterior en el intervalo $|y| < \delta$ y $J_N f(x_0)$ como la integral en el intervalo $\delta \leqslant |y| \leqslant 1/2$. Entonces $S_N f(x_0) - f(x_0) = I_N f(x_0) + J_N f(x_0)$.

$$\begin{aligned} J_N f(x_0) &= \int_{\delta \leqslant |y| \leqslant \frac{1}{2}} \left[f(x_0-y) - f(x_0)\right] \frac{e^{\pi i(2N+1)y} - e^{-\pi i (2N+1)y}}{2i\sin(\pi y)}\,\mathrm{d}y \\ &= \int_{-1/2}^{1/2} h_{+,x_0}(y) e^{2\pi i N y}\,\mathrm{d}y - \int_{-1/2}^{1/2} h_{-,x_0}(y) e^{-2\pi i N y}\,\mathrm{d}y, \end{aligned}$$

donde $h_{\pm,x_0}(y) = \left[f(x_0-y) - f(x_0)\right] \dfrac{e^{\pm\pi i y}}{2i\sin(\pi y)} \mathbb{1}_{\left\{\delta \leqslant |y| \leqslant 1/2\right\}}(y)$. Entonces, tenemos que

$$J_N f(x_0) = \widehat{h_{+,x_0}}(-N) - \widehat{h_{-,x_0}}(N) \xrightarrow[N\to\infty]{} 0$$

por el Lema de Riemann-Lebesgue. Solo falta ver que $h_{\pm,x_0} \in \mathcal{L}^1\left(\left[-1/2, 1/2\right]\right)$:

$$\int_{-\frac{1}{2}}^{\frac{1}{2}} |h_{\pm,x_0}(y)|\,\mathrm{d}y = \int_{\delta\leqslant|y|\leqslant\frac{1}{2}} |f(x_0-y) - f(x_0)| \frac{1}{2\,|\sin(\pi y)|}\,\mathrm{d}y.$$

Como $\forall y \in \left[-1/2, 1/2\right] : \delta \leqslant |y| \leqslant 1/2 \implies |\sin(\pi y)| \geqslant |\sin(\pi\delta)| > 0$, tenemos que

$$\|h_{\pm,x_0}\|_1 \leqslant \frac{1}{2\,|\sin(\pi\delta)|} \int_{-\frac{1}{2}}^{\frac{1}{2}} |f(x_0-y) - f(x_0)|\,\mathrm{d}y \leqslant \frac{1}{\cancel{2}\,|\sin(\pi\delta)|} \cancel{2}\,\|f\|_1 < \infty.$$

Por otro lado, para $I_N f(x_0)$, tenemos que

$$\begin{aligned}
I_N f(x_0) &= \int_{|y|<\delta} [f(x_0-y)-f(x_0)] \frac{e^{\pi i(2N+1)y}-e^{-\pi i(2N+1)y}}{2i\sin(\pi y)} \,\mathrm{d}y \\
&= \int_{-1/2}^{1/2} g_{+,x_0}(y) e^{2\pi i N y} \,\mathrm{d}y - \int_{-1/2}^{1/2} g_{-,x_0}(y) e^{-2\pi i N y} \,\mathrm{d}y,
\end{aligned}$$

donde $g_{\pm,x_0}(y) = [f(x_0-y)-f(x_0)] \dfrac{e^{\pm\pi i y}}{2i\sin(\pi y)} \mathbb{1}_{\{|y|<\delta\}}(y)$. Nuevamente, por el Lema de Riemann-Lebesgue, tenemos que $I_N f(x_0) \xrightarrow[N\to\infty]{} 0$ siempre que $g_{\pm,x_0} \in \mathcal{L}^1$. Pero esto es cierto ya que

$$\|g_{\pm,x_0}\|_1 = \int_{|y|<\delta} |f(x_0-y)-f(x_0)| \frac{1}{2|\sin(\pi y)|} \,\mathrm{d}y.$$

Como $\forall y \in (-\delta,\delta) : |\sin(\pi y)| \geqslant 2|y|$, tenemos que

$$\|g_{\pm,x_0}\|_1 \leqslant \int_{|y|<\delta} \frac{|f(x_0-y)-f(x_0)|}{4|y|} \,\mathrm{d}y.$$

Esta integral es finita ya que f satisface el criterio de Dini en x_0. ∎

Referenciado en

- `prop-clase-ck-velocidad-convergencia-uniforme-fourier`

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